Image Defogging using Dark Channel Prior and Optimization (DRIVE GUARDIAN)

Yousuf , Divyansh , Rohan , Shaurya
Bennett University

REPORT

1 Introduction

Images captured under outdoor conditions experience the influence of fog, haze, and other dust particles present in the air. It leads to poor image contrast because of light scattering.

Outdoor images have a lot of applications. For instance, they are used for the purposes of navigation and autonomous driving.

There exist multiple methods of outdoor image improvement. However, all of them lack the knowledge of the laws that cause atmospheric scattering. In spite of the fact that the method based on Dark Channel Prior shows better results than the others, it suffers from the problems that appear when the image contains sky regions or places of high pixel intensities.

2 Review of Existing Systems and Feasibility

2.1 Review of Existing Systems

Several algorithms have been suggested to address the problem of defogging images. Simple algorithms like Histogram Equalization enhance the contrast but they lack the consideration of the physics behind haze formation. The Dark Channel Prior method offers an enhanced solution based on some properties of the statistical nature of images; however, its results are poor in areas with high intensity such as sky due to artifacts produced in this process.

2.2 Feasibility Analysis

The feasibility of this solution lies in the fact that the algorithm employs simple and straightforward algorithms that are normally used in image processing. The algorithm is computationally cheap and can be employed in any normal computer without any need for special equipment.

3 Objective

One of the main objectives of the proposed method is to remove the fog from the image in order to increase its visibility. Moreover, it is aimed to improve image contrast while keeping

the edges of small objects intact.

Besides, it strives to eliminate the negative consequences of some methods. They produce halo-like artifacts in output images. Moreover, it aims to produce natural images.

4 Basic Concept

4.1 Atmospheric Model

When an image is captured in foggy conditions, there appears two factors in the image: the original scene and light scattering produced by fog. This process can be described with the help of an atmospheric scattering model.

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

Where:

- $I(x)$: foggy image
- $J(x)$: original scene
- $t(x)$: transmission (amount of light reaching the camera)
- A : atmospheric light (ambient light caused by scattering)

As per the model, an image is divided into two factors: the original scene and light scattering. They depend on the values of atmospheric transmission and atmospheric light correspondingly.

It allows understanding the influence of fog on images and gives a possibility to use mathematical approaches to create an improved image.

4.2 Dark Channel Prior

It utilizes the idea that in most cases outdoor images contain at least one color channel with the minimum pixel intensity.

$$D_{\text{dark}}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} J_c(y) \right) \tag{2}$$

Where:

- $D_{\text{dark}}(x)$: dark channel value
- $\Omega(x)$: local patch around pixel x
- $J_c(y)$: intensity of color channel c at pixel y
- $c \in \{R, G, B\}$: red, green, blue channels

However, this approach is problematic when there are sky or other places with high pixel intensities since there will be no very dark pixels in such images.

5 Proposed Method

5.1 Sky Segmentation

Division of the image into sky and non-sky regions contributes to increasing the effectiveness of the image processing procedure because of different influences of fog on these regions. Consequently, it makes it possible to prevent image distortions.

This approach makes it possible only with dividing the image into regions since different processing algorithms can be used for each region separately.

5.2 Estimating Atmospheric Light

Contrary to the algorithm based on Dark Channel Prior, it estimates atmospheric light for the sky and non-sky regions separately and then combines them into the final result.

$$A = W_{sky}A_{sky} + W_{non}A_{non} \tag{3}$$

Where:

- A_{sky} : atmospheric light for sky region
- A_{non} : atmospheric light for non-sky region
- W_{sky}, W_{non} : weights assigned to each region

This makes the procedure more accurate due to the fact that atmospheric light differs in the image.

5.3 Estimating Transmission

Similarly to the atmospheric light, transmission is calculated separately for the sky and non-sky regions.

$$t = q_1 t_{sky} + q_2 t_{non} \tag{4}$$

Where:

- t_{sky} : transmission in sky region
- t_{non} : transmission in non-sky region
- q_1, q_2 : weighting factors

Then, it is combined to get a single value. This is done taking into account the difference in fog amount in these regions.

5.4 Optimization

It assists the system in determining the best parameters for the algorithm. It is advantageous to perform a more accurate reconstruction since it does that on its own. This improves the quality of the output image because there is no need to set anything beforehand.

5.5 Enhancing the Image

After finding the necessary parameters, an image can be reconstructed according to the atmospheric scattering model.

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A \tag{5}$$

Where:

- t_0 : minimum transmission threshold to avoid division by zero

Next, it should be enhanced by adjustment of brightness and contrast levels to create a clear image.

5.6 Object Detection using YOLO

The suggested framework will also employ the YOLO model for detecting objects. The YOLO model is an object detection algorithm which plays an important role in recognizing regions of interest in the image.

The use of YOLO will help to have a better understanding of the image and will help to process the image efficiently during the defogging process.

6 AI Pipeline

The general process of the suggested system comprises several phases, which follow each other.

- Firstly, the fogged image serves as input data.
- Secondly, the YOLO algorithm helps identify relevant areas within the image.
- Further, the image is split into two classes – sky areas and non-sky areas.
- Atmospheric light estimation and transmission coefficients estimation are carried out separately for both classes.
- Next, the parameters are optimized.
- Finally, image reconstruction occurs based on the atmospheric scattering model.

7 System Flow Diagram

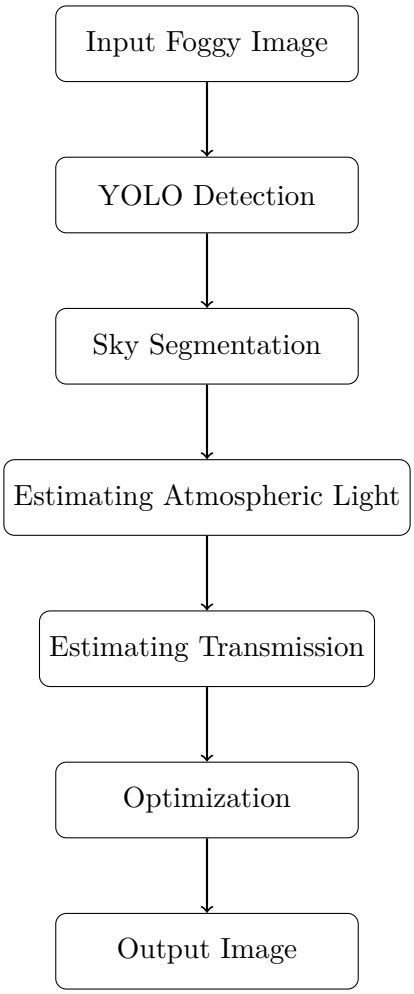

Figure 1: Flowchart of Image Defogging Process

8 Results and Discussion

Comparison of the output images produced by the proposed algorithm with those produced by similar methods proves the superior quality of the former images. They are sharper and brighter and therefore more visible.

Figure 2: Before fogged-up version (left) and After de-fogged version (right)

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9 Tech Stack

- Python
- OpenCV
- NumPy
- Matplotlib
- YOLO

10 References

[1] B. Li, R. Cao, H. Hu, and Z. Zhang, "Image defogging algorithm based on metaheuristic dark channel prior," *Signal Processing: Image Communication*, Elsevier.

11 Contribution of Each Team Member

- Yousuf – Worked on report writing and overall documentation
- Divyansh – Designed and created the project presentation (PPT)
- Rohan – Developed and implemented the project
- Shaurya – Worked on optimization and system flow design